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Gradient Descent.

- → Batch GD
- → Stochastic GD.
- → Mini Batch GD

Batch

- check all rows
- Slower

$\hat{y}_i = \beta_0 + \beta_1 x_1 + \beta_2 x_2$

x_1	x_2	y

Step 1: Start with random value.

Generally, $\underbrace{\beta_0 = 0}_{\text{intercept}}$, $\underbrace{\beta_1, \beta_2 = 1}_{\text{slope}}$

Step 2: epoch = 200, learning rate = 0.1.

$\beta_0 = \beta_0 - \eta \text{ slope} \rightarrow \frac{\partial L}{\partial \beta_0}$

$\beta_1 = \beta_1 - \eta \text{ slope} \rightarrow \frac{\partial L}{\partial \beta_1}$

$\beta_2 = \beta_2 - \eta \text{ slope} \rightarrow \frac{\partial L}{\partial \beta_2}$

~~Loss fn~~

Total ~~Loss~~ Square error: $L = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$

$\frac{\partial L}{\partial \beta_0}$

~~Beta~~ Mean Square error: $L = \frac{1}{n} \cdot \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$

n = total no. of rows.

For now let's take mean square root:

$$L = \frac{1}{2} \sum_{i=1}^{n=2} (y_i - \hat{y}_i)^2$$

$$= \frac{1}{2} \left[(y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2 \right]$$

$$\left\{ \begin{aligned} \hat{y} &= \beta_0 + \beta_1 x_1 + \beta_2 x_2 \\ \hat{y}_1 &= \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} \\ y_2 &= \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} \end{aligned} \right\}$$

$$L = \frac{1}{2} \left[(y_1 - \beta_0 - \beta_1 x_{11} - \beta_2 x_{12})^2 + (y_2 - \beta_0 - \beta_1 x_{21} - \beta_2 x_{22})^2 \right]$$

$$\frac{\partial L}{\partial \beta_0} = \frac{1}{2} \left[-2 (y_1 - \hat{y}_1) + (-2)(y_2 - \cancel{\beta_0 - \beta_2 x_{21}}\ \hat{y}_2) \right].$$

$$\frac{\partial L}{\partial \beta_0} = -\frac{2}{2} \left[(y_1 - \hat{y}_1) + (y_2 - \hat{y}_2) \right]$$

If $n = n$, then it would be (general)

$$\frac{\partial L}{\partial \beta_0} = -\frac{2}{n} \left[(y_1 - \hat{y}_1) + (y_2 - \hat{y}_2) + (y_3 - \hat{y}_3) + \cdots + (y_n - \hat{y}_n) \right]$$

$$\frac{\partial L}{\partial \beta_0} = -\frac{2}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)$$

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$$\frac{\partial L}{\partial \beta_1} = \frac{\partial}{\partial \beta_1}\left(\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2\right)$$

$$= \frac{\partial}{\partial \beta_1}\left(\frac{1}{n}\left[(y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2\right]\right)$$

$$= \frac{\partial}{\partial \beta_1}\left(\frac{1}{n}\left[(y_1 - \beta_0 - \beta_1 x_{11} - \beta_2 x_{12})^2 + (y_2 - \beta_0 - \beta_1 x_{21} - \beta_2 x_{22})^2\right]\right)$$

$$= \frac{1}{n}\left[2(y_1 - \hat{y}_1)(-x_{11}) + 2(y_2 - \hat{y}_2)(-x_{21})\right]$$

$$\frac{\partial L}{\partial \beta_1} = \frac{-2}{n}\left[(y_1 - \hat{y}_1)(x_{11}) + (y_2 - \hat{y}_2)(x_{21})\right]$$

$$= \frac{-2}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)\, x_{i1}$$

$x_{i1} \rightarrow 1^{st}$ col'n data.

11
21
31
41
51
⋮
n1.

$$\frac{\partial L}{\partial \beta_2} = \frac{-2}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)\, x_{i2}$$

⋮

$$\boxed{\frac{\partial L}{\partial \beta_m} = \frac{-2}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)\, x_{im}}$$

$y_i \rightarrow$ training data. output column

$\hat{y}_i \rightarrow$ predicted data

$(n \times m)$

x_1	x_2	y_1

$\hat{y}_i = \beta_0 + \beta_1 x_1$

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→ intercept_der = -2 * np.mean(y_train - y_hat)

$\hat{y}_i = \beta_0 + \beta_1 x_{11} + \beta_2 x_{12}$

$\hat{y}_1 = \beta_0 + [x_{11} \ x_{12}] \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}$

$\hat{y}_i$ = np.dot(coef_, X_train) + β_0.

→ for all:

y_hat $\hat{y}_i = \beta_0 + $ np.dot(X_train, coef_);

(353, 10) (10, 2)

(353, 2)

coefficient:

self.coef = self.coef_ - (self.lr * coef_der)

together

coef_der = -2 * np.dot((y_train - y_hat), X_train) / X_train.shape[0].

Batch → n = 1000

col → 5 → 6 total col.

epoch → 50 → 10^3.

1 → 1000 } × 50

6 → 6000

{50000} iteration.

Very slow on big datasets.

→ takes more RAM